ZigBee Network Technology

A ZigBee network, in a home or industry, can easily link electronic devices to save energy and money while protecting all electronic devices in the network. Home automation application will be able to monitor the amount of energy that users consume through ZigBee modules that will be receiving power signals from daily used devices such as lamp, radiator, ventilation, and others.

This smart meter connected to a ZigBee module every monitors and combines all electronic devices to form a large and complex network within a home or industry. Through this network, the smart meter will shout any device off when needed as well as when too much energy is used. The user will be able to manage their consumption conveniently and in detail. In fact, ZigBee network technology will not only make life more comfortable but also save energy and money while protecting your environment.

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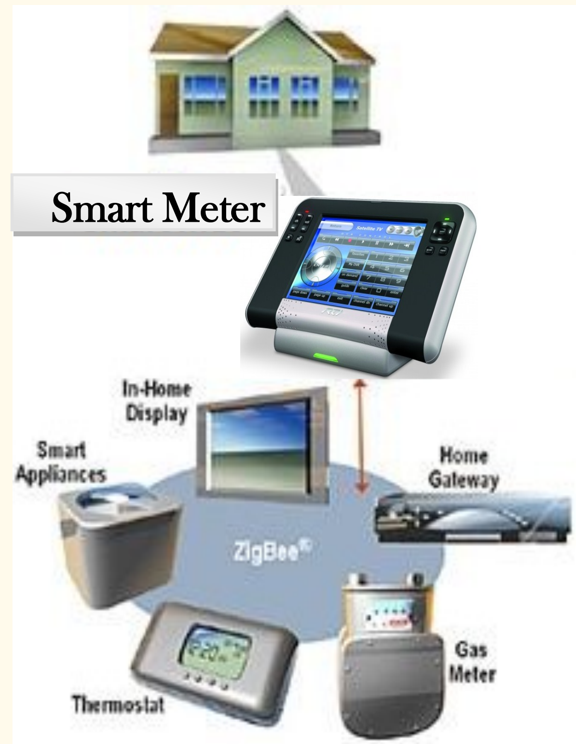
Smart Meter Application & Service Information

By Piter Garcia



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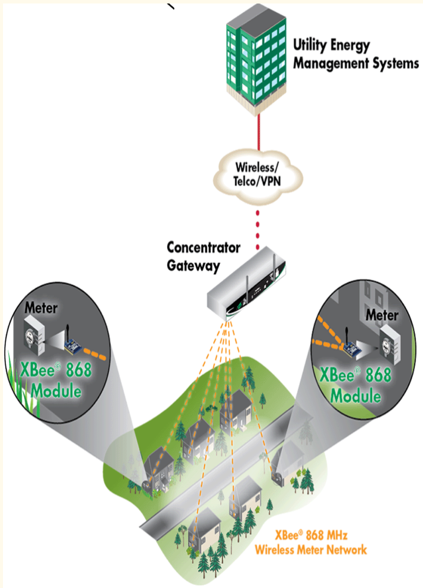
Smart Meter New Technology

The Smart Meter uses ZigBee protocols built in low power consume technology. All applications consume the least possible amount of energy to maintain all synchronized electronic devices within the same network in sleep mode.

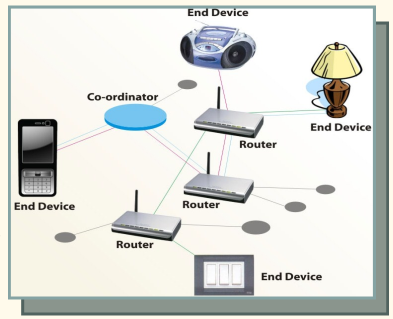


This new designed is a low-cost and low-data rate wireless technology that consist of three main devices to incorporate ZigBee radios application. A coordinator to organize the network and maintains routing tables. Routers to talk to the coordinator, routers and reduced-function end devices. And finally reduced-function end-devices that talk to routers and the coordinator, but not to each other.

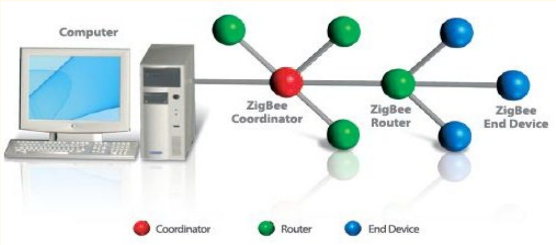
Routers and coordinator are mains-powered and do not go to sleep, nevertheless end-devices wake up only when they need to communicate and immediately go back to sleep to minimize power consumption and promote long battery

Affordable Technology

The wireless smart meter application is a cheap and easy way to provide network communication and power control between electronic devices at places with no wired infrastructure . The smart meter wireless communication provides all suitable data rates as compared to other wireless technology so this withstands suitably for all applications. This new technology also provides reliable protection for the operation of electric power systems and benefits to industrial automation applications because of low cost deployment and redeployment.



The smart meter network, when placed into our daily used devices such as lamp, radiator, ventilation and others, monitors by judging from the read out from the sensor the amount of energy consumed these appliances and shut them off when too much energy is being used. Therefore, devices like those mentioned above could easily be combined together to form a large and simple network within a home or a industry to make life more comfortable while saving energy. This smart meter application will for fact make it possible for users to conserve electricity and save money.



Smart Meter Network Functionality

ZigBee network layer supports different application methods such as star, tree, and mesh topologies. In the star topology, the network is controlled by one single device called the ZigBee coordinator. The ZigBee coordinator initiates the communication and maintain end-devices on the network. In tree networks, routers move data and control messages through the network using a hierarchical routing strategy. And in the mesh and tree topology the ZigBee coordinator starts the network and chooses certain key network parameters, network may be extended through the use of ZigBee routers. Mesh networks perform a complete peak-to-peak communication and it is described only by intra-PAN networks through which communications start and end within the same network.

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